Remote sensing and GIS technical portfolio

MD. NAHID TANBIR AHMED

Phone: +880 1987985043

Email: www.nahid.tanbir08@gmail.com

Linkedin: https://www.linkedin.com/in/nta08

Project Title : Drone Image Processing

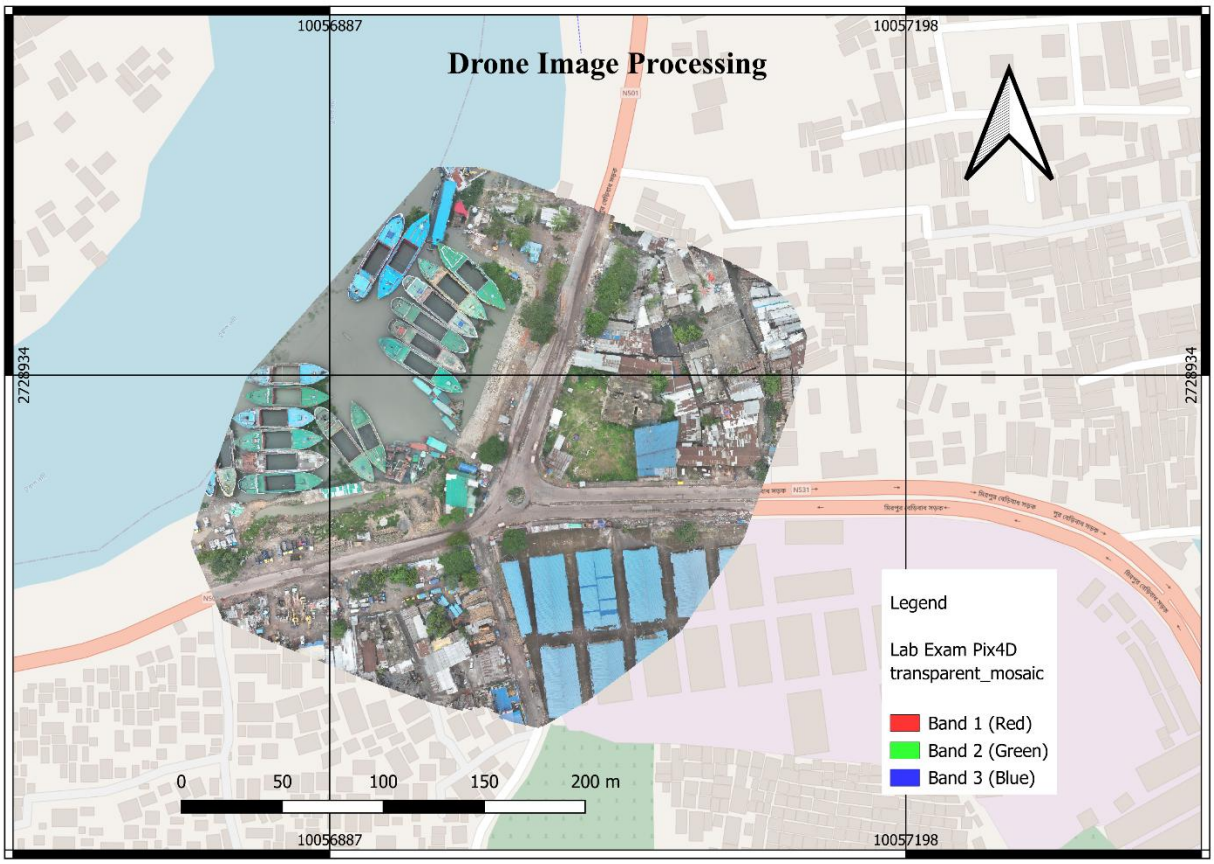
Study Area : Tongi (Nearby Turag River)

Tools: Pix4D mapper software, QGIS

Objective: To create a high resolution imagery

Data: Raw Image captured from Drone (DJI Mavic 2)

Output: converts overlapping imagery into Georeferenced orthomosaics, elevation models (DSM/DTM), dense 3D point clouds, textured meshes, and analytical measurement products.



Project Title: Identifying wetlands within a buffer zone (3km) and clip to extract wetland areas.

Study Area : Khulna Division

Tools: QGIS

Data: LGED, BWDB

Objective & Output: Performed proximity and area-specific analysis by creating buffers around roads and clipping spatial layers to the study area boundary. Resulting layers were used for flood risk mapping, exposure assessment, and focused spatial analysis.

Method:

- Created buffer zones around key features (roads) using the Buffer tool with specified distances.
- Extracted study area–specific data by clipping larger spatial layers using the Clip tool.
- Prepared resulting layers for overlay and zonal analysis in GIS.

Preview:

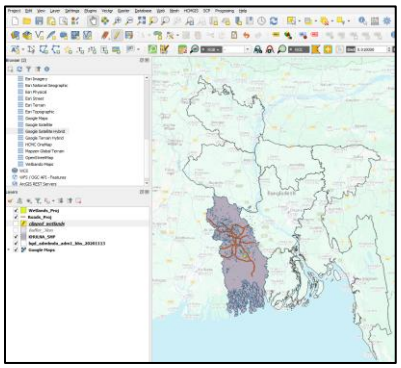


Fig: Data Input

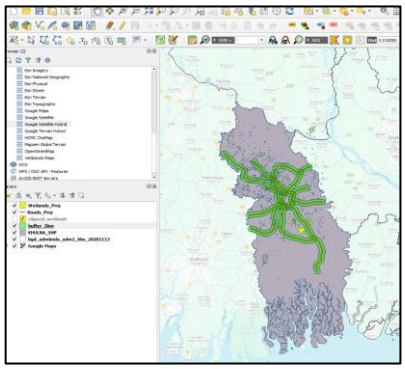


Fig: Creating Buffer (3km)

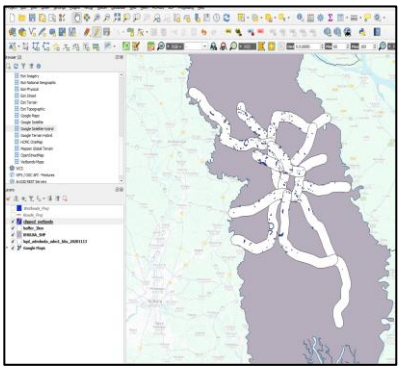


Fig: Clipped Wetlands within the buffer zone

Project Title: An overall survey on different types of shops in University Premise.

Study Area: Jahangirnagar University

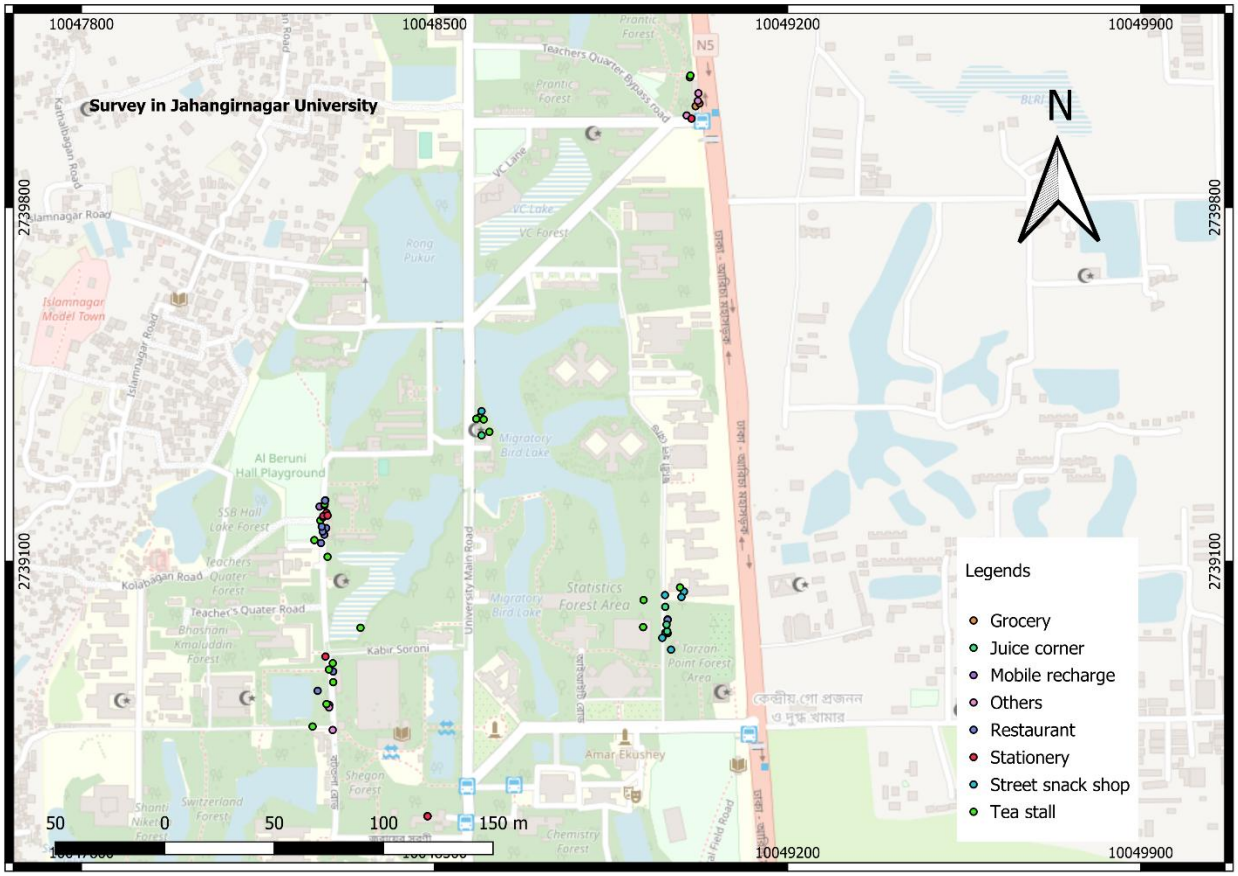
Tools: Kobo Toolbox, Microsoft Excell, QGIS

Objective: To make an overall understanding about university area's shops and stalls and visualization

Data: Collected by field survey

Method: Individual questioning, observing.

Output: Making a map with the details of shops category and its location.



Project Title: Dizitization of Gauripur Upazila

Study Area : Gauripur (Barisal Division)

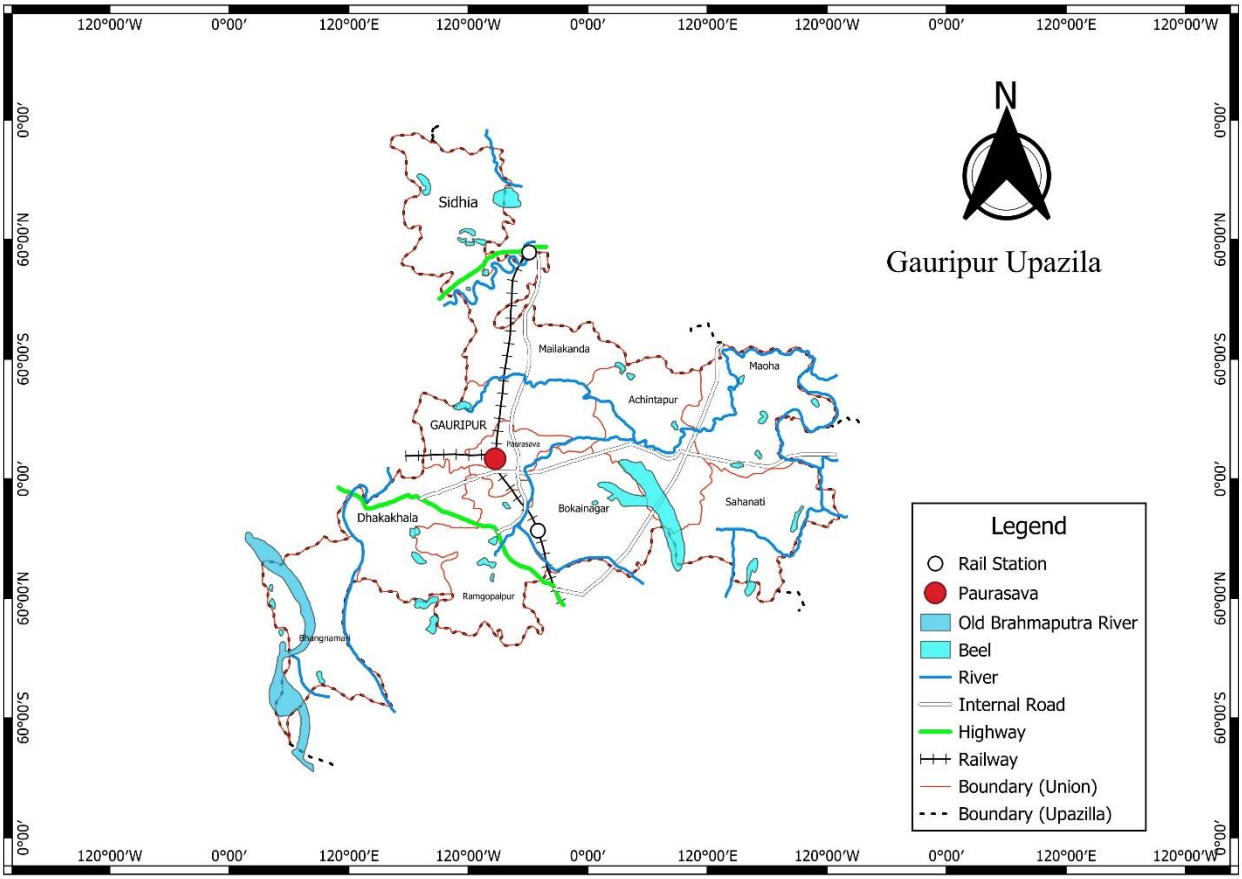
Tools: QGIS

Objective: Convert raster-based visual information into structured, analyzable, and attribute-linked vector data for spatial analysis and decision-making.

Data: Collected from internet.

Method: Using digitization features in QGIS

Output: Making a map with proper notations, symbology, and details.



Project Title: Making a Population intensity map of Dhaka (South-East region)

Study Area: Dhaka Division

Tools: ArcGIS Pro and QGIS

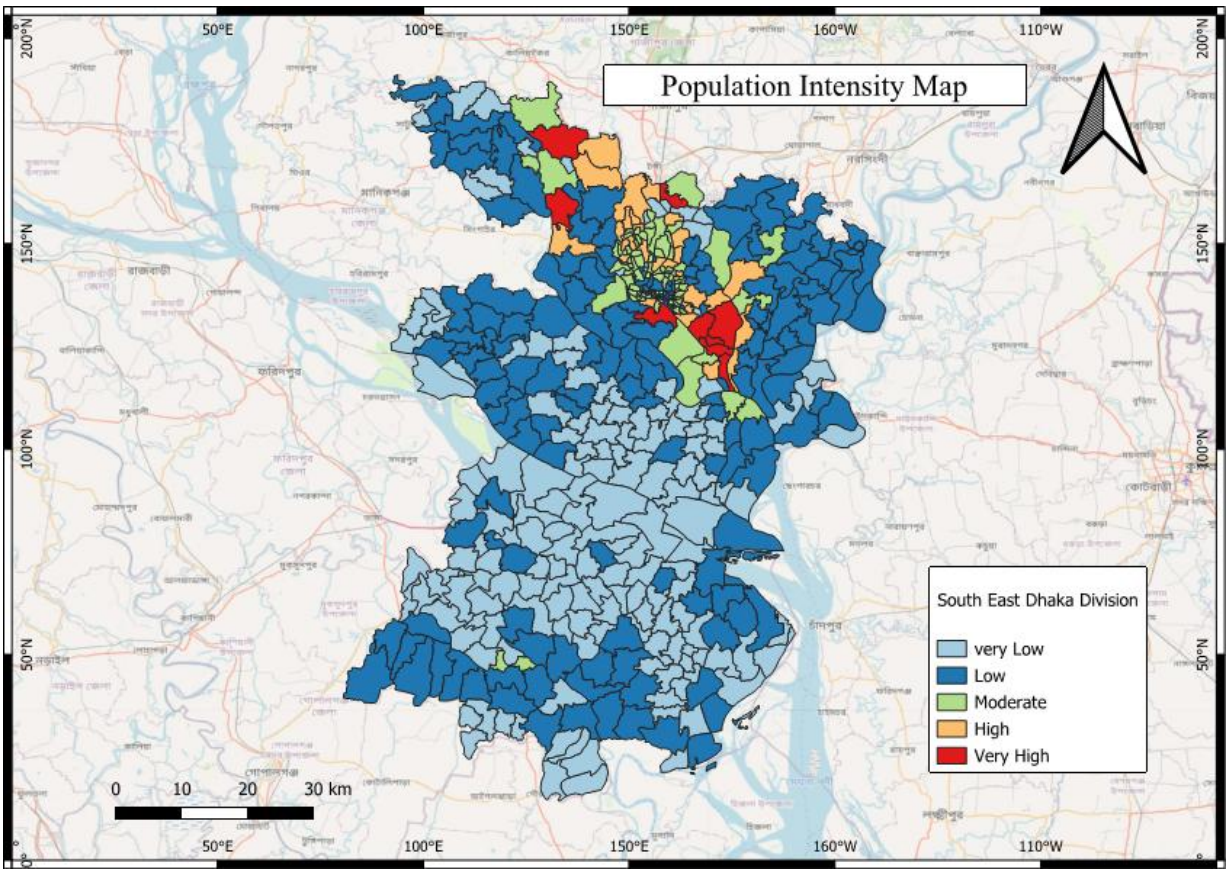
Objective: The objective of a population intensity map is to represent the spatial concentration of people per unit area in order to support planning, infrastructure management, and disaster risk assessment.

Data: Census data + boundary shapefile.

Method: Area calculation → Population/Area ratio → Graduated classification

Output: Highest intensity recorded in central urban zones

Preview:



Project Title: LULC classification and Accuracy Assessment Using Supervised Classification of Cahndpur District

Study Area: Chandpur District

Tools: ERDAS imagine, ArcGIS, QGIS, Microsoft Excell

Objective: To produce and validate a Random Forest–based LULC classification map using Landsat imagery.

Data: LandSAT 8

Method: Random Forest classification → Accuracy assessment (Overall, Producer, User, Kappa)

Output: Classified map with 92% overall accuracy.

Preview:

LULC Classification and Accuracy Assessment Using Supervised Classification in ArcGIS

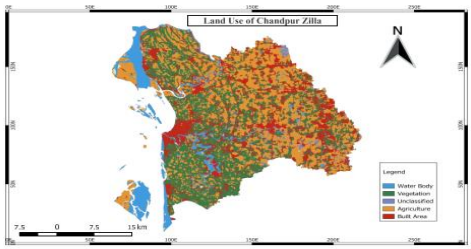


Fig: LULC map of Chandpur District

Fig: Ground Truth Points

	OBJECTID*	ClassVALUE	C_0	C_1	C_2	C_3	C_4	C_5	Total	U_Accuracy	Kappa
2	1	C_0	10	0	0	0	0	0	10	1	0
3	2	C_1	0	9	0	1	0	0	10	0.9	0
4	3	C_2	0	1	8	1	0	0	10	0.9	0
5	4	C_3	0	0	1	9	0	0	10	0.9	0
6	5	C_4	0	0	0	0	9	1	10	0.9	0
7	6	C_5	0	0	0	0	0	0	0	0	0
8	7	Total	10	10	9	10	9	1	50	0	0
9	8	P_Accuracy	1	0.8	0.9	0.9	1	0	0	0.92	0
10	9	Kappa	0	0		0	0	0	0	0	0.8923861

Fig: Accuracy Assessment Table

This table is a confusion matrix used to assess the accuracy of a classified image (e.g., from remote sensing land cover classification). It compares the reference (ground truth) data against the classified output for five land cover classes: Waterbody, Vegetation, Agriculture, Barren Land, and Urban Area.

Class	Ground	Correctly Classified	Misclassified	User Accuracy	Producer Accuracy
C_0	Waterbody	10	0	**1.00 (100%)**	**1.00 (100%)**
C_1	Vegetation	9	1 (to Agriculture)	**0.9 (90%)**	**0.9 (90%)**
C_2	Agriculture	8	1 (from Vegetation) + 1 (from Urban)	**0.8 (80%)**	**0.9 (90%)**
C_3	Barren Land	9	1 (to Urban)	**0.9 (90%)**	**0.9 (90%)**
C_4	Urban Area	9	1 (from Barren Land)	**0.9 (90%)**	**0.9 (90%)**

At a glance:

Overall Accuracy: 92%
Average Producer's Accuracy: 92%
Average User's Accuracy: 92%
Kappa Coefficient: 0.8924

Conclusion:

The confusion matrix demonstrates that the land cover classification performed with high accuracy. The overall accuracy is 92% and the Kappa coefficient is 0.892, indicating a strong correlation between classified results and ground truth data. While most classes such as Waterbody and Urban Area were classified with high precision, minor confusion was observed between Vegetation and Agriculture, as well as between Barren Land and Urban Area

Project Title: Radiometric correction of Satellite image

Objective: Correct satellite imagery to remove sensor and atmospheric distortions and improve data reliability for subsequent analysis.

Tools: ERDAS imagine software

Data: Landsat 8

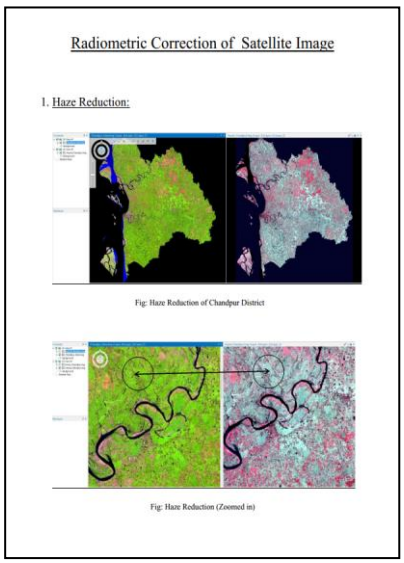
Method:

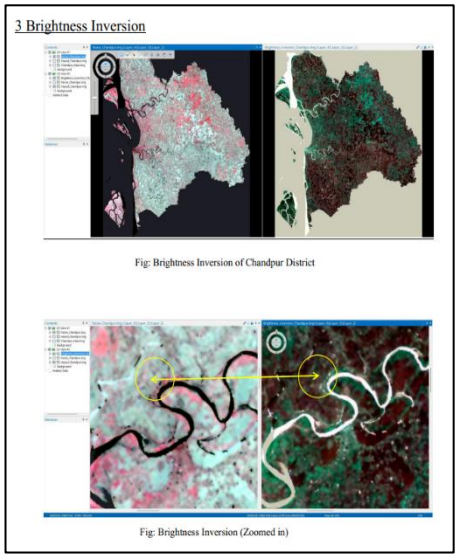
- Radiometric calibration to convert DN to reflectance
- Atmospheric correction to minimize scattering and absorption effects
- Enhanced image quality for accurate feature extraction

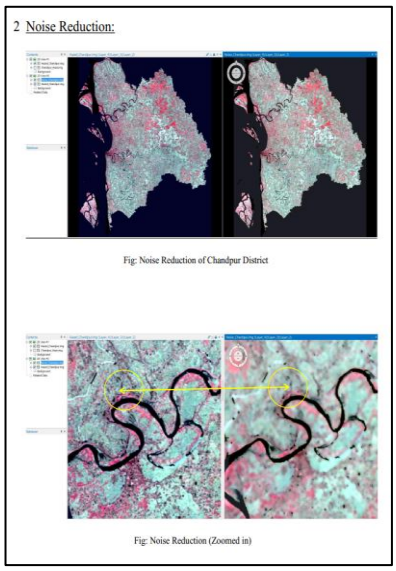
Output:

- Corrected satellite imagery with consistent reflectance values
- Ready for NDWI, LULC, or change detection analysis

Preview:







Project Title: Population Data Analysis Using Python

Tools: Python, Jupyter Notebook
Data: population.csv (district-level voters and population)

Objective: Analyze district-level population data to compute descriptive statistics and extract insights using array/dataframe operations.

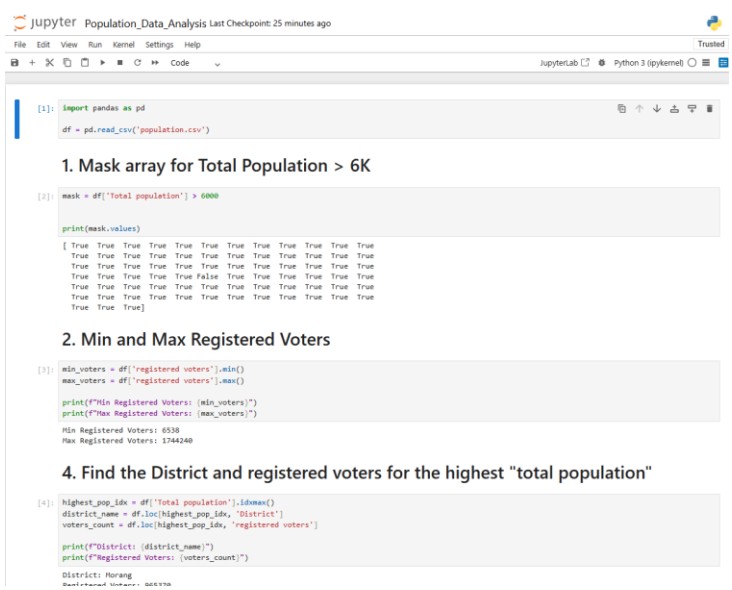
Method:

- Applied filtering (masking) to identify districts with population thresholds
- Used aggregation functions (min, max, mean) for statistical analysis
- Retrieved specific district records through indexing

Key Outputs:

- Minimum & maximum registered voters identified
- District with lowest population detected
- District with highest population extracted
- Average population calculated for the dataset
- Subset of first 15 districts analyzed

Code Preview:



GitHub Repository: https://github.com/nahidtanbir08/population-data-analysis.git

(Full notebook, dataset, and outputs available in the repository.)